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Quantum Machine Learning and Quantum Amplitude Estimation for Short-Horizon Index Option Trading: An Empirical Study on NSE Tick Data

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ABSTRACT Quantum computing is often proposed for two finance tasks: classifying market data with quantum machine learning, and pricing derivatives with quantum amplitude estimation (QAE). We put both to the test on real data from the National Stock Exchange of India (NSE). Using 69,519,707 recorded ticks of NIFTY and BANKNIFTY options over 36 sessions, we label each candidate option purchase by its profit after buying at the ask, selling at the bid 180 seconds later, and paying all brokerage, taxes and fees. We then benchmark a fidelity quantum kernel SVM, a projected quantum kernel SVM and a data re-uploading variational quantum classifier with classical models on the same reduced features, using walk-forward validation by trading day. We simulate every quantum model exactly and verify the simulators against Qiskit and PennyLane. The best quantum model reaches an out-of-sample AUC of 0.535, below classical models on the same inputs (up to 0.556) and below gradient boosting on all features (0.588); No filter, quantum or classical, makes a profit after costs. We also measure kernel concentration, barren plateaus, shot noise and depolarizing noise. For pricing, iterative QAE prices 24 recorded contracts within 0.76% of the exact discretized value and shows the expected quadratic speedup over Monte Carlo, with error falling as $N^{-0.86}$ against $N^{-0.50}$. Circuit counts and noisy simulations tell a harder story: on current hardware, this advantage remains out of reach. Today's quantum methods leave the economics of short-horizon option buying unchanged.

INDEX TERMS Amplitude estimation, financial markets, NSE, option pricing, quantum computing, quantum kernel methods, quantum machine learning, variational quantum circuits.

I. INTRODUCTION

QUANTUM computing is widely expected to change computational finance [1]–[3]. Two applications dominate the discussion. The first is quantum machine learning (QML), where data are mapped into the exponentially large state space of a quantum register in the hope of finding patterns that classical models miss [4]–[6]. The second is derivative pricing with quantum amplitude estimation (QAE), which estimates an expected payoff with error $O(1/N)$ in the number of queries N, a quadratic improvement over the $O(1/\sqrt{N})$ error of classical Monte Carlo [7]–[10].

Most published tests of these ideas rely on synthetic data, small textbook problems or daily prices. Real high-frequency market data are a different matter: signals are weak, costs are large, and every claim must survive an out-of-sample check. Few studies have gone there. Index options on the National Stock Exchange of India (NSE) are a demanding test case. They are among the most traded derivatives in the world [11], and a SEBI study found that 93% of individual traders in equity derivatives lost money between FY22 and FY24, with total losses above Rs 1.8 lakh crore [12]. Much of that loss is mechanical. Every round trip pays the bid–ask spread, brokerage and statutory charges, and over a few minutes these costs dwarf the typical price move.

We ask two concrete questions of recorded NSE tick data. First, can a quantum classifier, used as a filter before each trade, select option purchases that are profitable after all costs better than a classical model can? Second, how well does QAE price the same contracts, and how far is its theoretical speedup from practical use? We answer both with a design

built to resist optimistic results: labels from future tradable prices after costs, walk-forward validation by trading day, identical inputs for quantum and classical models, and tests that account for comparing many models [13]–[16].

This paper makes four contributions.

1) We benchmark three quantum classifiers (fidelity quantum kernel, projected quantum kernel and a data re-uploading variational classifier) like for like against classical models on the same reduced features, the same training samples and the same walk-forward folds, and we evaluate them on 125,703 out-of-sample trades with both statistical and after-cost profit measures.
2) We quantify the quantum effects that limit these models in practice: kernel concentration as the number of qubits grows, barren plateaus in variational training, finite measurement shots and depolarizing gate noise.
3) We use QAE to price 24 recorded NIFTY and BANKNIFTY options; we separate discretization, payoff-approximation and estimation errors, measure the convergence rate against classical Monte Carlo, count circuit resources and simulate hardware noise.
4) We release exact state-vector simulators for the feature map and the variational circuit, validated against Qiskit and PennyLane, together with an open pipeline that regenerates every number in the paper from the recorded data.

The answer to the first question is no. Given the same information, no quantum classifier beats the classical models, and no model of either kind turns a profit after costs. The second answer is more nuanced: QAE behaves exactly as theory predicts on a simulator, yet it demands deep, noise-sensitive circuits, and even a perfect price would leave untouched the trading costs that drive the losses.

Section II introduces the quantum methods. Section III reviews related work. Sections IV and V describe the data, the labels and the validation design. Section VI details the quantum models and experiments. Section VII reports the results, Section VIII discusses them, and Section IX concludes.

II. QUANTUM COMPUTING BACKGROUND

A. QUBITS, GATES AND MEASUREMENT

The state of an n-qubit register is a unit vector in $\mathbb{C}^{2^n}$,

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i \, |i\rangle \,, \qquad \sum_i |\alpha_i|^2 = 1, \tag{1}$$

where $|i\rangle$ are the computational basis states [17]. Computation applies unitary gates such as the Hadamard gate H, single-qubit rotations $R_Y(\theta) = e^{-i\theta Y/2}$ and $R_Z(\theta) = e^{-i\theta Z/2}$, and the two-qubit CNOT. Measuring the register returns basis state $|i\rangle$ with probability $|\alpha_i|^2$, so expectation values such as $\langle Z_0 \rangle = \langle\psi| \, Z_0 \, |\psi\rangle$ must be estimated from repeated runs, or shots. Current devices have tens to a few hundred noisy qubits and limited circuit depth, the noisy intermediate-scale quantum (NISQ) regime [18].

B. QUANTUM KERNEL METHODS

A quantum kernel method encodes a data point $\mathbf{x} \in \mathbb{R}^n$ in a state $|\phi(\mathbf{x})\rangle = U(\mathbf{x}) \, |0\rangle^{\otimes n}$ and uses the overlap of two states as a kernel [5], [6]. We use the second-order Pauli-Z (ZZ) feature map of Havlíček et al. [5],

$$U(\mathbf{x}) = \left(U_\Phi(\mathbf{x}) H^{\otimes n}\right)^2, \tag{2}$$

$$U_\Phi(\mathbf{x}) = \exp\Big(i \sum_k x_k Z_k + i \sum_{k<l} (\pi - x_k)(\pi - x_l) Z_k Z_l\Big), \tag{3}$$

whose circuit is shown in Fig. 1(a). The fidelity kernel is

$$k_F(\mathbf{x}, \mathbf{x}') = |\langle \phi(\mathbf{x}) | \phi(\mathbf{x}') \rangle|^2, \tag{4}$$

which a quantum computer estimates as the probability of measuring $|0\rangle^{\otimes n}$ after applying $U(\mathbf{x})$ and then $U^\dagger(\mathbf{x}')$. Inputs are scaled by a bandwidth λ, $\mathbf{x} \mapsto \lambda \mathbf{x}$, which strongly affects performance [19]. The projected quantum kernel of Huang et al. [20] instead compares the one-qubit reduced density matrices $\rho_k(\mathbf{x})$,

$$k_P(\mathbf{x}, \mathbf{x}') = \exp\Big(-\gamma \sum_{k=1}^{n} \big\| \rho_k(\mathbf{x}) - \rho_k(\mathbf{x}') \big\|_F^2\Big). \tag{5}$$

Because $\|\rho - \rho'\|_F^2 = \frac{1}{2} \sum_{P \in \{X,Y,Z\}} (\langle P \rangle - \langle P \rangle')^2$ for one qubit, k_P is a Gaussian kernel on the $3n$ single-qubit Pauli expectations. Both kernels are used in a support vector machine (SVM) [21].

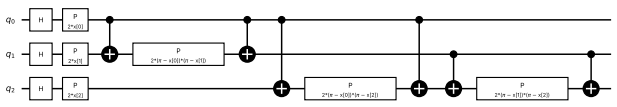
Quantum kernels share a known weakness, exponential concentration. For expressive feature maps, kernel values for different inputs approach a constant as n grows, and their variance shrinks exponentially; telling them apart then takes an exponential number of shots [22]. The kernel goes blind.

C. VARIATIONAL QUANTUM CLASSIFIERS

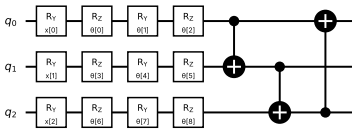
A variational quantum classifier uses a parameterized circuit as a trainable model [23]. We use data re-uploading [24], in which the input is encoded again before every trainable block:

$$|\psi_{\boldsymbol{\theta}}(\mathbf{x})\rangle = \prod_{l=1}^{L} W_l(\boldsymbol{\theta}_l) \, S(\mathbf{x}) \, |0\rangle^{\otimes n}, \quad S(\mathbf{x}) = \bigotimes_{k=1}^{n} R_Y(x_k), \tag{6}$$

where each block W_l applies a general rotation $R_Z(\omega) R_Y(\theta) R_Z(\phi)$ to every qubit followed by a ring of CNOT gates whose range changes from block to block (Fig. 1(b)). The model output is $f(\mathbf{x}) = a \langle Z_0 \rangle_{\boldsymbol{\theta}} + b$, and the predicted probability of a profitable trade is $\sigma(f(\mathbf{x}))$. Training minimizes the logistic loss with gradient descent. For randomly initialized circuits, the variance of the gradient can decay exponentially with n. Training then stalls. These barren plateaus remain the central obstacle for variational methods [23], [25].



(a) ZZ feature map, one repetition, three qubits



(b) One block of the variational classifier

FIGURE 1. Quantum circuits used for classification, drawn with Qiskit for three qubits. The experiments use 6 qubits, two repetitions of (a) and four blocks of (b).

D. QUANTUM AMPLITUDE ESTIMATION FOR OPTION PRICING

Let $\mathcal{A}$ be a circuit that loads the risk-neutral distribution of the index at expiry onto n qubits and encodes the option payoff in an extra qubit:

$$\mathcal{A}\,|0\rangle^{\otimes(n+1)} = \sum_{i=0}^{2^n-1} \sqrt{p_i}\;|i\rangle\left(\sqrt{1-\tilde{f}(x_i)}\,|0\rangle + \sqrt{\tilde{f}(x_i)}\,|1\rangle\right), \tag{7}$$

where x_i are 2^n grid points of the index level, p_i their probabilities and $\tilde{f}(x) \in [0,1]$ an encoding of the payoff. The probability of measuring $|1\rangle$ in the last qubit is $a = \sum_i p_i \tilde{f}(x_i)$. Following Stamatopoulos et al. [10] and Woerner and Egger [26], we use $\tilde{f}(x) = \sin^2\left(c\,\hat{f}(x) + \pi/4\right)$, where $\hat{f} \in [-1,1]$ is the normalized payoff and c a rescaling factor. For small c, $\tilde{f} \approx \frac{1}{2} + c\hat{f}$, so the expected payoff follows linearly from a, with an approximation error that shrinks as c decreases.

QAE estimates $a = \sin^2\theta_a$ using the Grover operator

$$\mathcal{Q} = \mathcal{A}S_0\mathcal{A}^\dagger S_{\psi_0}, \tag{8}$$

which rotates the state by $2\theta_a$; S_{ψ_0} flips the sign of the good states and S_0 reflects about the initial state [7]. Applying $\mathcal{Q}^k$ amplifies the dependence of the measurement outcome on θ_a, which gives an estimation error of order $1/N$ after N applications of $\mathcal{A}$, compared with $1/\sqrt{N}$ for N classical samples [8]. We use Iterative Amplitude Estimation (IAE) [27], which avoids the quantum Fourier transform and reaches a target accuracy ϵ with confidence $1-\alpha$ using only Grover powers and classical post-processing, a variant related to amplitude estimation without phase estimation [28]. The distribution is loaded with a state-preparation circuit for a discretized log-normal law [29], [30].

III. RELATED WORK

A. QUANTUM COMPUTING IN FINANCE

Surveys by Orús et al. [1], Egger et al. [2] and Herman et al. [3] identify optimization, machine learning and Monte Carlo pricing as the main candidate applications. Rebentrost et al. [9] first described quantum Monte Carlo pricing of derivatives, Woerner and Egger [26] applied QAE to risk measures, and Stamatopoulos et al. [10] gave circuits for European, basket and path-dependent options and ran small instances on IBM hardware. Chakrabarti et al. [31] estimated that a practical quantum advantage in derivative pricing requires thousands of error-corrected logical qubits and very deep circuits. All of these studies price contracts from assumed model parameters. We price contracts taken directly from NSE tick data.

B. QUANTUM MACHINE LEARNING

Quantum kernel methods were introduced by Havlíček et al. [5] and Schuld and Killoran [6]. Huang et al. [20] showed that classical models trained on data can match quantum models unless the quantum kernel is carefully designed, and proposed projected kernels. Shaydulin and Wild [19] showed the importance of the kernel bandwidth, and Thanasilp et al. [22] proved that many quantum kernels concentrate exponentially. Variational classifiers [23], [24] face barren plateaus [25]. A large benchmark by Bowles et al. [16] found that out-of-the-box classical models usually outperform quantum classifiers on standard datasets. We extend this evidence to real, noisy, cost-sensitive market data, where the target is profit after costs, not accuracy.

C. CLASSICAL MACHINE LEARNING AND MARKET MICROSTRUCTURE

Short-horizon prices are driven by order flow and liquidity [32]–[35], and machine learning on order-book data mostly predicts mid-price direction [36]–[38]. Comparisons of tree ensembles and neural networks for return prediction find that gains shrink after costs [39], [40]. Random or overlapping train–test splits overstate performance on financial time series [13], [41], [42], and comparing many models requires corrections such as White's Reality Check and Hansen's SPA test [14], [15], [43]. We hold quantum and classical models to the same safeguards.

IV. MARKET SETTING AND DATA

A. NSE INDEX OPTIONS AND TRADING COSTS

NIFTY and BANKNIFTY options are European, cash-settled contracts on the NIFTY 50 and NIFTY Bank indices. Premiums are quoted in rupees with a tick size of Rs 0.05. Each contract covers one lot. After the revision of lot sizes announced in October 2025, a lot is 65 units for NIFTY and 30 units for BANKNIFTY for contracts expiring after December 30, 2025 [44]. Since November 2024, weekly expiries are permitted for only one benchmark index per exchange, which on NSE is NIFTY [45].

A round trip in one lot incurs the charges listed in Table 1. Brokerage is a flat fee per order at discount brokers. The exchange transaction charge, the SEBI fee and stamp duty are proportional to premium turnover. STT is levied on the sale premium, and GST applies to brokerage, the exchange

TABLE 1. Transaction Cost Schedule Applied to Each Round Trip

Component	Base	Rate
Brokerage	Per order	Rs 20 (two orders)
Exchange transaction charge	Buy + sell premium	0.03553% [46]
SEBI turnover fee	Buy + sell premium	Rs 10 per crore
Stamp duty	Buy premium	0.003%
STT	Sell premium	0.15% (0.10% before Apr. 1, 2026) [47]
GST	Brokerage + exchange + SEBI fee	18%
Bid–ask spread	Entry and exit quotes	Observed

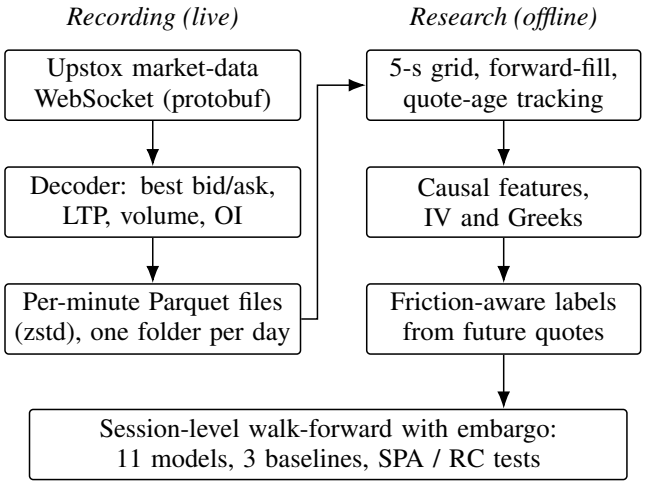


FIGURE 2. Data recording and research pipeline. Only information time-stamped at or before a decision time is used to build features for that decision.

charge and the SEBI fee. Our cost model applies the rate in force on each trade date.

B. DATA COLLECTION SYSTEM

Fig. 2 summarizes the recording system we built. Our recorder process subscribes to the Upstox market-data WebSocket, which delivers protocol buffer messages for a set of NIFTY and BANKNIFTY option contracts around the at-the-money strike and for the two indices. The subscription is refreshed as the at-the-money strike moves. Each message is decoded into a tick record containing the last traded price, best bid and ask, cumulative traded volume, open interest, the exchange's last-trade time and the local receive time. During the sample period the subscription covered the 15 strikes nearest the money for each index, both calls and puts, in the nearest NIFTY weekly expiry and the nearest BANKNIFTY monthly expiry. Ticks are buffered in memory and written once per minute to Parquet files with Zstandard compression, giving one directory per trading day. We run all analysis offline from these files with the Polars dataframe library and scikit-learn [48].

Two properties of the feed matter for the analysis. First, the exchange timestamp carried by each tick is the time of the last trade, not of the last quote update. We therefore compute feed latency only on ticks that carry a new trade time. The resulting receive-minus-trade delay has a median of 257 ms and a 95th percentile of 11,355 ms. This is an upper bound on transport latency, because it also includes any clock offset between the exchange and the recording machine. The long upper tail comes from thinly traded strikes, for which a snapshot message can arrive several seconds after the trade it reports. Second, the recorder stores only the best bid and ask for each contract, so queue sizes and deeper levels of the book are not available.

TABLE 2. Candidate Trades at the Primary Horizon

Underlying	Events	Calls (%)	Net-profitable (%)	Median spread (%)
BANKNIFTY	117,437	50.1	30.0	0.33
NIFTY	113,735	49.9	30.3	0.26
All	231,172	50.0	30.2	0.29

C. CLEANING AND SAMPLING

The recorder captured 41 weekday sessions with option data between June and October 2026. A session enters the study only if candidate trades exist in at least 180 distinct minutes of the 375-minute trading day. We fixed this rule before fitting any model; it removes 5 sessions in which the recorder ran for only part of the day. The remaining 36 sessions, from June 19, 2026 to August 18, 2026, contain 69,519,707 ticks in 56,188 files. We keep ticks received between 09:15 and 15:30 Indian Standard Time and marked as live data, and remove exact duplicates. Quotes with a non-positive bid or an ask below the bid are treated as missing, and so are zero values of volume and open interest, which the feed uses when a field is absent from a message. For each contract, the last observation in every 5-second interval is placed on a regular grid and carried forward. The age of the last real update is kept so that stale quotes can be excluded. The index level comes from the subscribed index instruments. If those are unavailable at a grid point, it is recovered from put–call parity at the strike with the smallest call–put price gap.

Candidate trades are taken on a grid aligned to the holding horizon h, so that successive candidates in the same contract do not overlap. A candidate requires a live quote no older than 30 seconds, an ask of at least Rs 2, a relative spread below 50%, an entry at least five minutes after the open, and enough time to the close and to expiry to complete the holding period. A candidate is dropped if no quote younger than 30 seconds exists at the exit time. Table 2 describes the resulting sample at the primary horizon of 180 s.

V. LABELS, FEATURES AND VALIDATION

A. FRICTION-AWARE LABELS

Let a candidate purchase of contract i be considered at time t, with best ask a_t, best bid b_t, mid-quote $m_t = (a_t + b_t)/2$ and lot size q. The trade buys one lot at a_t and sells it at the best bid b_{t+h} observed h seconds later. With buy and sell turnover

TABLE 3. Feature Set (29 Features in Six Groups)

Group	Features
Liquidity	Relative spread $(a_t - b_t)/m_t$; spread in ticks; age of the last quote update
Flow	Traded volume over 1 and 5 min; OI velocity and acceleration (12); VWOIM (13)
Momentum	Option mid log returns over 1 and 5 min; index log returns over 1 and 5 min; 5-min realized index volatility
Greeks	Implied volatility; $\lvert\Delta\rvert$; $\Gamma_{\%}$ (16); vanna; $\ln(K/S)$; minutes to expiry; IV minus at-the-money IV
Market	Put–call OI ratio and its change; net gamma share (17); at-the-money IV and its change
Context	Minute of session; expiry-day flag; call flag; BANKNIFTY flag

$B = qa_t$ and $S = qb_{t+h}$, the total charge is

$$C = 2c_{\mathrm{b}} + (\phi_{\mathrm{x}} + \phi_{\mathrm{s}})(B + S) + \phi_{\mathrm{d}} B + \phi_{\mathrm{t}} S + g\left[2c_{\mathrm{b}} + (\phi_{\mathrm{x}} + \phi_{\mathrm{s}})(B + S)\right], \tag{9}$$

where c_{b} is brokerage per order, ϕ_{x}, ϕ_{s}, ϕ_{d} and ϕ_{t} are the exchange, SEBI, stamp duty and STT rates, and g is the GST rate (Table 1). The realized net profit and the label are

$$\pi_{i,t} = S - B - C, \qquad y_{i,t} = \mathbb{1}\{\pi_{i,t} > 0\}. \tag{10}$$

The label depends only on quotes after t and on the published cost schedule. None of the model inputs is used to construct it. We deliberately fix the trade mechanics: every candidate is entered and exited with marketable orders, holds one lot, and exits after exactly h seconds. We do not model limit-order entries, stop-loss and trailing-stop exits, profit targets, longer holding periods or position sizing. The results therefore answer a narrow question: can a gate rescue this simple, common form of short-horizon option buying? They say nothing about whether some other rule-based system built on the same data could profit.

The net profit decomposes exactly into a mid-quote move, a spread-crossing cost and fees:

$$\pi_{i,t} = \underbrace{q(m_{t+h} - m_t)}_{\text{mid move}} - \underbrace{q\left[(a_t - m_t) + (m_{t+h} - b_{t+h})\right]}_{\text{spread crossing}} - C. \tag{11}$$

Section VII uses (11) to trace where the gross move goes.

B. FEATURES

Table 3 lists the 29 features, grouped into six families. Every feature at time t is computed from grid values at or before t within the same session. Windows of one and five minutes correspond to 12 and 60 grid steps.

1) Open-interest flow

Let OI_t be open interest and $\Delta\mathrm{OI}_t^{(w)} = \mathrm{OI}_t - \mathrm{OI}_{t-w}$. Over a one-minute window w_1 we define velocity and acceleration as

$$v_t = \frac{\Delta\mathrm{OI}_t^{(w_1)}}{\mathrm{OI}_{t-w_1}}, \qquad \alpha_t = v_t - v_{t-w_1}. \tag{12}$$

Rising prices with rising open interest indicate new long positions, while rising prices with falling open interest indicate short covering. To capture this interaction we define the volume-weighted open-interest momentum

$$\mathrm{VWOIM}_t = \mathrm{sgn}(m_t - \overline{m}_t)\, \ln\!\left(1 + V_t^{(w_5)}\right) \frac{\Delta\mathrm{OI}_t^{(w_5)}}{\mathrm{OI}_{t-w_5}}, \tag{13}$$

where $\overline{m}_t$ is the session volume-weighted average mid-quote up to t and $V_t^{(w_5)}$ is the volume traded over the last five minutes.

2) Implied volatility and Greeks

NSE index options are European, so we invert the Black–Scholes formula [49], [50] at the mid-quote by bisection on $\sigma \in [0.01, 5]$. Time to expiry τ is measured in years to 15:30 on the expiry date, and the risk-free rate is held constant because its effect over minutes is negligible. With

$$d_1 = \frac{\ln(S/K) + (r + \tfrac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}, \tag{14}$$

and φ the standard normal density, we use

$$\Gamma = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}}, \qquad \mathrm{Vanna} = \frac{\partial^2 V}{\partial S\, \partial\sigma} = -\varphi(d_1)\frac{d_2}{\sigma}. \tag{15}$$

Gamma is reported as the change in delta, in option units, for a 1% move of the index,

$$\Gamma_{\%} = 0.01\, \Gamma S^2. \tag{16}$$

Contracts whose mid-quote violates the no-arbitrage bound have no implied volatility and are left missing. The models handle these values through imputation or native missing-value support.

3) Market context

Dealer positions are not observable. Following the convention used in gamma-exposure proxies, we assume that customers are net buyers of puts and net sellers of calls, so that dealers are long call gamma and short put gamma [51]. The net gamma share across the subscribed strikes of an underlying is

$$\mathrm{NGS}_t = \frac{\sum_{\text{calls}} \Gamma_{\%}\,\mathrm{OI} - \sum_{\text{puts}} \Gamma_{\%}\,\mathrm{OI}}{\sum_{\text{all}} \Gamma_{\%}\,\mathrm{OI}} \in [-1, 1]. \tag{17}$$

The put–call ratio is the ratio of put to call open interest over the same strikes.

4) Time-of-day normalization

Volume, spreads and open-interest changes follow a strong intraday pattern. For the six features most affected (both volume windows, OI velocity and acceleration, VWOIM and the relative spread), each value is standardized within its 15-minute slot as

$$z_t = \mathrm{clip}\left(\frac{x_t - \mathrm{med}_{s(t)}}{\mathrm{IQR}_{s(t)} + \epsilon}, -10, 10\right), \tag{18}$$

where the median and interquartile range for slot $s(t)$ are estimated on the training sessions of each fold only.

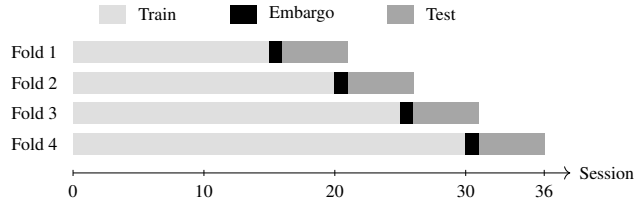


FIGURE 3. Expanding-window walk-forward design over the 36 trading sessions. Each training window is further split into an inner fitting part and a validation part that sets the decision threshold.

C. VALIDATION DESIGN

Fig. 3 illustrates the walk-forward scheme. Sessions are ordered by date. The first fold trains on the first 15 sessions, skips one embargo session, and tests on the next five. Each later fold adds five sessions to the training window, which gives 4 folds and 20 test sessions. Because all labels end within the session in which they start, splitting by session already prevents label overlap. The embargo session adds protection against dependence that persists overnight, for example in open interest.

Within each training window, the last quarter of the sessions, after a further embargo session, serves as an inner validation set. A model is first fitted on the earlier part and used to choose the decision threshold on the validation part. It is then refitted on the full training window and applied to the test block with that threshold. The threshold maximizes total net profit on the validation sessions, subject to at least 30 validation trades. If no threshold yields a positive total, the model abstains for the whole test block. Not trading is always an option. Time-of-day statistics, imputation medians and scaling parameters are likewise estimated on training data only.

VI. QUANTUM MODELS AND EXPERIMENTAL DESIGN

A. INPUTS FOR THE QUANTUM MODELS

Simulating a quantum model is expensive, and a NISQ device has few qubits. We therefore describe each candidate trade with just 6 numbers. Within each training window, the 29 features of Table 3 are imputed with training medians, standardized, reduced by principal component analysis (PCA) to 6 components, standardized again and clipped to $[-3, 3]$. We train each quantum model on a random sample of 3,000 training events and give the same sample and inputs to three classical models: logistic regression, an RBF-kernel SVM and a small multilayer perceptron (MLP) with 8 hidden units, which has about as many parameters as the variational classifier (74). Gradient boosting on all 29 features and all training events, the strongest model of the classical study, serves as a reference. Every out-of-sample trade receives a test prediction.

B. MODEL SETTINGS

The fidelity kernel uses the ZZ map (2) with full entanglement and two repetitions. The projected kernel uses the same states with γ set by the median heuristic. The bandwidth $\lambda \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$ of each kernel is chosen in every fold by the AUC on inner validation sessions, and the SVM uses $C = 1$ with Platt scaling [52]. The variational classifier (6) has 6 qubits and four blocks. Its input scale is chosen from $\{0.25, 0.5, 1\}$ on validation data, and it is trained for 40 epochs with Adam (learning rate 0.05, batch size 256). Decision thresholds are chosen on validation sessions to maximize profit, exactly as in Section V.

C. SIMULATION AND VALIDATION

We simulate all quantum models exactly. For the ZZ map we compute state vectors directly: $H^{\otimes n}$ is applied with a fast Walsh–Hadamard transform and U_Φ is diagonal in the computational basis, which makes it possible to simulate hundreds of thousands of states. We simulate the variational circuit in JAX with automatic differentiation and check both simulators against circuit-level references: the ZZ states agree with Qiskit [53] to a fidelity error below 10^{-14}, and $\langle Z_0 \rangle$ of the variational circuit agrees with PennyLane [54] to 10^{-15}.

D. HARDWARE EFFECTS

We study three effects of real hardware. Qubit count: the PCA dimension, and hence the number of qubits, is varied from 2 to 10. Shot noise: each kernel entry is replaced by $\hat{k} = B/S$ with $B \sim \mathrm{Binomial}(S, k)$, as when k is estimated from S shots. Gate noise: under global depolarizing noise, the compute–uncompute circuit for k_F returns

$$k_{\text{noisy}} = f\, k_F + (1 - f)\, 2^{-n}, \qquad f = (1 - p_2)^{N_{\text{CX}}}, \quad (19)$$

where p_2 is the CNOT error rate and $N_{\text{CX}} = 120$ is the number of CNOT gates for 6 qubits. We measure concentration by the variance of off-diagonal kernel values on 400 inputs for 2 to 14 qubits, and barren plateaus by the variance of $\partial \langle Z_0 \rangle / \partial \theta$ over 100 random initializations for 2 to 12 qubits.

E. QUANTUM AMPLITUDE ESTIMATION SETUP

We apply QAE to 24 contracts recorded at 11:00 on three test sessions: for each of NIFTY and BANKNIFTY, the two calls and two puts nearest the money. Spot, strike, implied volatility and time to expiry come from the recorded data. The index at expiry is modeled as log-normal under the risk-neutral measure, discretized on 2^n points over the mean ± 3 standard deviations, with $n = 5$ and $c = 0.05$. IAE uses target accuracy $\epsilon = 10^{-3}$, confidence level $\alpha = 0.05$ and 1,000 shots per circuit. We report the Black–Scholes (BS) price, the exact price on the grid, the QAE estimate and the market mid-quote. We measure convergence on one at-the-money NIFTY call by repeating IAE 20 times for each ϵ and comparing the root-mean-square error (RMSE) of $\hat{a}$ with that of classical Monte Carlo using the same number of samples. Circuit resources are counted after transpiling $\mathcal{A}$ and $\mathcal{Q}$ to CNOT and single-qubit gates, and noise is simulated with Qiskit Aer using depolarizing errors on every gate.

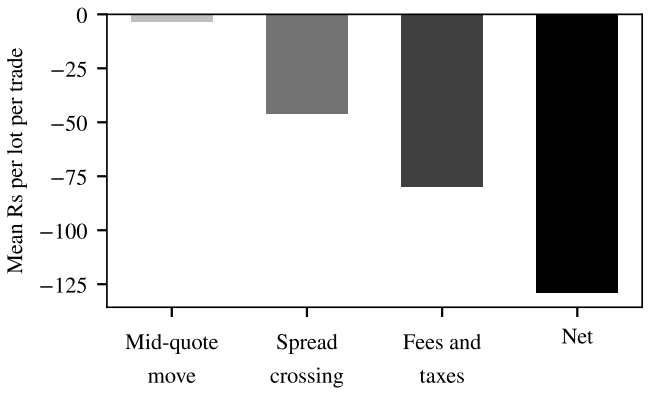


FIGURE 4. Average decomposition of net profit per lot for all out-of-sample candidate trades, following (11).

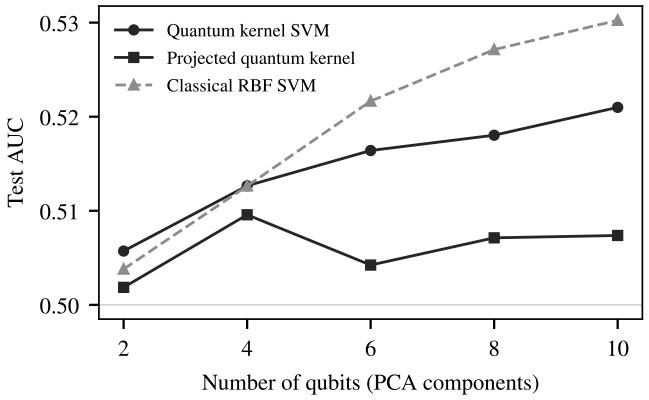


FIGURE 5. Test AUC against the number of qubits (equal to the number of PCA inputs), on 10,000 test trades per fold.

VII. RESULTS

Unless stated otherwise, results refer to the holding period of 180 s and the 125,703 out-of-sample trades over 20 test sessions.

A. THE COST HURDLE

Fig. 4 splits the result of every test trade using (11). The average mid-price move per lot is Rs −3.3. The spread costs Rs 45.9 on average, and fees and taxes cost Rs 80.0. Buying every candidate gives an average result of Rs −129.2 per trade, and only 30.1% of these trades make money. The price move is close to zero, and slightly negative, because a bought option bleeds time value. The costs, about Rs 126 per lot, do not trim a profitable trade. They are the entire loss. Fees and taxes make up about two-thirds of it. The largest single item is brokerage with GST, a fixed Rs 47.20 per round trip, which hurts most on cheap contracts. The spread is smaller because the near-the-money contracts in our sample are liquid: the median spread is 0.26% of the price for NIFTY and 0.33% for BANKNIFTY (Table 2). This steady cost drag fits with the large losses of individual traders reported by SEBI [12].

B. QUANTUM AND CLASSICAL CLASSIFIERS

Table 4 compares all models. The fidelity quantum kernel reaches an AUC of 0.514 and the projected quantum kernel 0.506, barely above the no-skill value of 0.5. The variational quantum classifier does better, with 0.535, but it is still below logistic regression (0.547) and the small MLP (0.556) trained on exactly the same inputs and samples. Gradient boosting on all features reaches 0.588. The AUC of the best quantum model minus that of the best classical model is -0.053, with a 95% session-bootstrap confidence interval of [-0.070, -0.034], so the gap is statistically clear. Every model posts a Brier score close to the 0.210 of a constant forecast. The signal is weak.

The economics tell the same story. On most validation sets no threshold produced a profit, so the two quantum kernel models and gradient boosting never traded. The variational classifier traded 7,550 times and lost Rs −128.6 per trade on average, about the same as buying every candidate. Even the top tenth of each model's predictions loses between Rs 95 and Rs 128 per lot. Across all models, the Reality Check and the SPA test against not trading give p-values of 1.000 and 1.000.

C. SCALING WITH THE NUMBER OF QUBITS

Fig. 5 varies the number of qubits. The fidelity kernel improves slightly with more qubits, but it stays below the classical RBF kernel on the same inputs at every size, and the projected kernel stays close to 0.5. More qubits mean more PCA components, so this curve mixes the effect of extra information with the effect of a larger quantum state. Even with that help, the quantum kernels trail.

D. KERNEL CONCENTRATION AND BARREN PLATEAUS

Fig. 6(a) shows exponential concentration of the fidelity kernel. With $\lambda = 1$, the variance of off-diagonal kernel values falls from 5.4×10^{-2} at 2 qubits to 9.5×10^{-9} at 14 qubits, about 1.90 bits per added qubit. A smaller bandwidth slows the decay, which is why small values of λ were selected on validation data. Fig. 6(b) shows the barren plateau of the variational circuit: the gradient variance falls from 1.0×10^{-1} at 2 qubits to 1.0×10^{-4} at 12 qubits, about 1.01 bits per qubit. Adding qubits, in short, makes these models harder to use, not better.

E. SHOT NOISE AND GATE NOISE

Table 5 shows the fidelity kernel under finite shots and depolarizing noise, on the same 10,000 test trades per fold as the exact reference (AUC 0.516). Neither effect changes the AUC by more than the sampling variation. Do not read this as robustness. The noise-free model already sits close to chance, so little signal remains to destroy. With $p_2 = 0.01$, a typical two-qubit error rate today, the damping factor f is about 0.3, so more than two-thirds of the kernel signal would be replaced by a constant on real hardware.

TABLE 4. Out-of-Sample Comparison of Quantum and Classical Classifiers

Type	Model	AUC	Brier	Top-10% hit	Top-10% mean	Trades	Mean/trade
Quantum	Quantum kernel SVM (fidelity)	0.514	0.211	33.1%	−118.8	0	–
Quantum	Projected quantum kernel SVM	0.506	0.210	31.8%	−127.9	0	–
Quantum	Variational quantum classifier	0.535	0.211	36.0%	−120.5	7,550	−128.6
Classical	Logistic regression	0.547	0.210	35.1%	−128.8	2,517	−136.0
Classical	Classical RBF SVM	0.522	0.210	34.8%	−105.1	399	−68.7
Classical	Classical MLP (8 units)	0.556	0.210	36.5%	−128.0	2,227	−123.6
Classical	Gradient boosting (29 features, all data)*	0.588	0.207	39.3%	−95.6	0	–

All models except * use the same 6 PCA inputs and the same 3,000 training events in each fold. Top-10%: the tenth of trades with the highest predicted probability. Trades and mean per trade (Rs per lot) refer to the thresholds chosen on validation sessions; "–" means the model never traded.

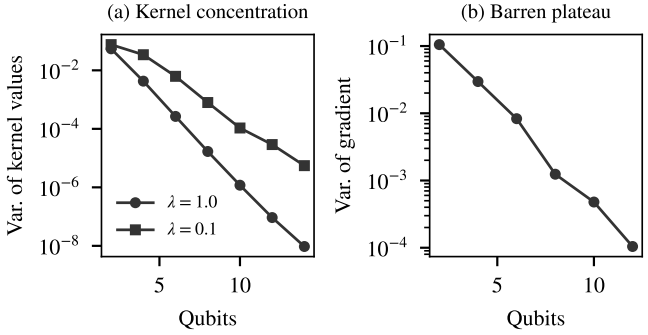


FIGURE 6. (a) Variance of fidelity-kernel values between different inputs. (b) Variance of the gradient of the variational circuit at random initialization. Both decay exponentially with the number of qubits.

TABLE 5. Fidelity Kernel Under Simulated Hardware Effects

Kernel estimate	Damping f	Test AUC
Exact (state vector)	1.000	0.516
100 shots, noiseless	1.000	0.512
1,000 shots, noiseless	1.000	0.517
10,000 shots, noiseless	1.000	0.516
1,000 shots, $p_2 = 0.001$	0.887	0.509
1,000 shots, $p_2 = 0.005$	0.548	0.521
1,000 shots, $p_2 = 0.01$	0.299	0.508

f: depolarizing damping factor (19). AUC on 10,000 test trades per fold.

F. QAE PRICING OF RECORDED CONTRACTS

Table 6 prices the 24 recorded contracts. The QAE estimates differ from the exact grid prices by a median of 0.76% (maximum 3.25%), using a median of 66,000 oracle queries per contract. With 5 qubits for the index, the grid itself differs from the BS price by a median of 0.7%. Because the implied volatility of each contract is computed from its own mid-quote, the BS price matches the market by construction, and the QAE price differs from the market by a median of 0.8%, the sum of the discretization and estimation errors. QAE reproduces the model price it is given. It cannot improve on the model.

Fig. 7 separates the two approximation errors. The discretization error falls quickly with the number of index qubits (a), and the payoff approximation error falls as c decreases (b). A smaller c, however, makes the useful part of the amplitude smaller, so more queries are needed for the same price accuracy.

TABLE 6. QAE Prices of Recorded NSE Options (Rs)

Index	Type	K	τ (d)	Market	BS	Grid	QAE	Queries
NIFTY	CE	24,150	5.2	157.23	157.22	156.07	157.18	67,000
NIFTY	CE	24,200	5.2	130.45	130.45	129.56	131.64	56,000
NIFTY	PE	24,150	5.2	113.55	113.55	112.85	114.64	76,000
NIFTY	PE	24,200	5.2	135.60	135.60	135.17	135.60	59,000
BANKNIFTY	CE	57,800	12.2	765.55	765.55	757.64	747.72	60,000
BANKNIFTY	CE	57,900	12.2	707.70	707.70	701.52	702.81	93,000
BANKNIFTY	PE	57,800	12.2	556.78	556.77	553.80	554.69	60,000
BANKNIFTY	PE	57,900	12.2	600.27	600.27	599.15	603.45	132,000
NIFTY	CE	24,250	5.2	102.45	102.45	101.70	100.51	60,000
NIFTY	CE	24,300	5.2	79.20	79.20	78.30	78.98	86,000
NIFTY	PE	24,250	5.2	121.55	121.55	120.71	120.99	86,000
NIFTY	PE	24,300	5.2	148.75	148.75	148.03	149.38	62,000
BANKNIFTY	CE	56,900	26.2	885.50	885.50	875.82	874.26	89,000
BANKNIFTY	CE	56,800	26.2	940.38	940.37	932.74	937.28	68,000
BANKNIFTY	PE	56,900	26.2	785.47	785.47	780.90	775.02	81,000
BANKNIFTY	PE	56,800	26.2	741.35	741.35	739.59	745.92	87,000
NIFTY	CE	24,300	6.2	173.75	173.75	172.25	172.94	65,000
NIFTY	CE	24,350	6.2	145.27	145.28	144.26	147.37	63,000
NIFTY	PE	24,300	6.2	103.55	103.55	102.96	101.06	62,000
NIFTY	PE	24,350	6.2	125.20	125.20	124.78	123.83	58,000
BANKNIFTY	CE	57,600	13.2	650.75	650.75	643.78	638.20	116,000
BANKNIFTY	CE	57,700	13.2	596.47	596.47	591.66	610.87	58,000
BANKNIFTY	PE	57,600	13.2	466.15	466.15	462.57	461.39	83,000
BANKNIFTY	PE	57,700	13.2	509.57	509.57	508.35	508.21	59,000

Market: recorded mid-quote. BS: Black–Scholes with the recorded implied volatility. Grid: exact expected payoff on the 2^5-point grid. QAE: iterative amplitude estimation. Queries: oracle calls to $\mathcal{A}$.

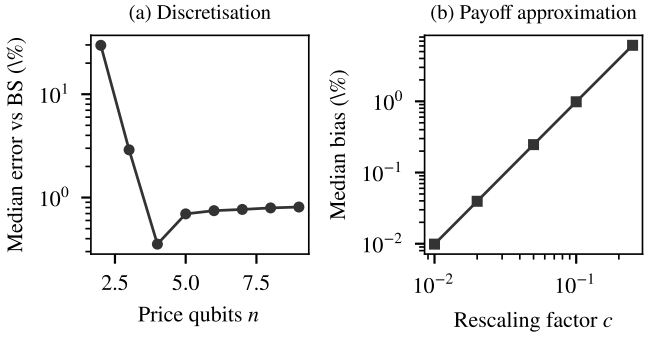


FIGURE 7. (a) Median error of the grid price against BS as the number of index qubits grows. (b) Median bias from the payoff approximation as a function of the rescaling factor c.

G. QAE CONVERGENCE AND COST

Fig. 8 confirms the quadratic speedup on a simulator. The RMSE of IAE falls as $N^{-0.86}$ with the number of oracle queries, against $N^{-0.50}$ for classical Monte Carlo; at about 3,000 queries IAE is 8.3 times more accurate. The catch: this advantage is counted in calls to $\mathcal{A}$, and each call is a

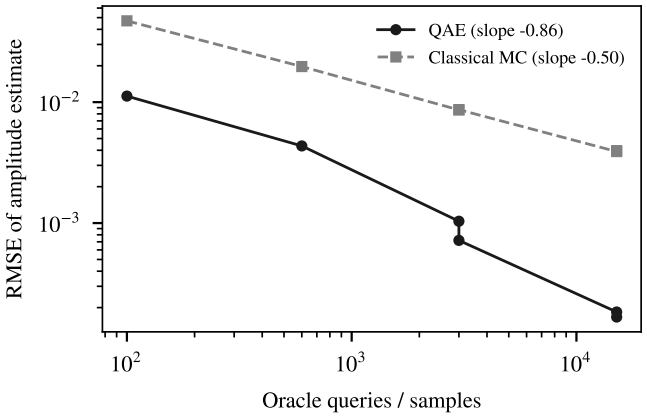


FIGURE 8. RMSE of the amplitude estimate against the number of oracle queries (QAE) or samples (classical Monte Carlo) for an at-the-money NIFTY call, 20 repetitions per point.

TABLE 7. Circuit Resources After Transpilation

n	Qubits	CX in $\mathcal{A}$	Depth $\mathcal{A}$	CX in $\mathcal{Q}$	Depth $\mathcal{Q}$
3	7	90	132	302	382
4	9	137	193	500	544
5	11	196	303	754	802
6	13	267	423	1,064	1,087
7	15	370	610	1,470	1,506
8	17	537	918	2,036	2,167

n: index qubits. $\mathcal{A}$: distribution loading and payoff. $\mathcal{Q}$: one Grover operator. Gates counted after transpiling to CNOT and single-qubit gates.

TABLE 8. IAE Under Depolarizing Noise (3 Index Qubits)

CX error p_2	Mean $\lvert\hat{a}-a\rvert$	Price error
0	0.0012	2.0%
0.001	0.0501	84.0%
0.005	0.1338	224.3%
0.01	0.1345	225.5%

Single-qubit error rate set to $p_2/10$. Price error relative to the exact grid price; mean of five runs.

deep circuit. Table 7 shows that one Grover operator already needs 302 CNOT gates at 3 index qubits and 2,036 at 8, and IAE applies it many times in sequence. Table 8 shows that depolarizing errors of 0.1% to 1% per CNOT already produce price errors far larger than the statistical error of the noise-free algorithm. A practical advantage would need error-corrected hardware, in line with the resource estimates of Chakrabarti et al. [31].

VIII. DISCUSSION

A. WHAT THE RESULTS MEAN

For classification, the quantum models found nothing the classical models missed. Given the same 6 inputs, logistic regression and a small MLP did as well as or better than every quantum model, and a classical model with all features did better still. This agrees with the large benchmark of Bowles et al. [16] and with the analysis of Huang et al. [20], which shows that data can let classical models match quantum kernels. The weak signal in this problem is mostly about which contracts suffer least from fixed costs, a simple, nearly monotone pattern that does not need a high-dimensional quantum feature space. Concentration and barren plateaus show that scaling these models to more qubits would make them harder to train and to estimate. Bigger is not better here.

For pricing, QAE delivers what theory promises: with exact simulation it needs far fewer queries than Monte Carlo for the same accuracy. But the advantage depends on running deep circuits without errors. On present hardware the noise would dominate, and for a single European option the closed-form BS price is already instant. For traders, the bigger point is that accurate pricing does nothing about the losses we measure. The costs of about Rs 126 per lot come from the spread, brokerage and taxes, which no pricing algorithm changes.

B. LIMITATIONS

Our study has five limitations. First, all quantum models are simulated; no experiment was run on quantum hardware, and noise is represented by simple models. Second, the quantum classifiers use 6 PCA components and 3,000 training events per fold, which is typical of current QML studies but discards information that the gradient boosting reference uses; better encodings or larger devices could change the comparison. Third, only one feature map and one variational architecture are tested, with a small hyperparameter search. Fourth, QAE prices European options under a log-normal model with one volatility, so it inherits the model's mismatch with market prices. Fifth, the data cover 36 sessions in one market period, and every trade uses market orders and a fixed holding time, as in the classical study.

IX. CONCLUSION

We tested two leading quantum computing methods on recorded NSE index-option data, and neither changed the outcome for the trader. Three quantum classifiers, compared like for like with classical models, did not improve the selection of profitable trades, and no model, quantum or classical, made money after costs. Quantum kernels concentrated and variational gradients vanished as qubits were added, and shot and gate noise would further weaken the models on real devices. QAE priced recorded contracts accurately on a simulator and showed the expected quadratic speedup in queries, but its circuits are deep, noise quickly dominates, and a better price does not reduce trading costs. Next, we plan to run the same benchmarks on quantum hardware, test problem-specific feature maps and quantum-native data, and apply QAE to path-dependent products where classical pricing is harder.

DATA AND CODE AVAILABILITY

The code for the classical and quantum experiments, including the validated quantum simulators, is available at https://github.com/harsh01103/stockezee. The raw tick data were obtained through a broker market-data interface whose terms do not permit redistribution; they are available from the author on reasonable request, subject to those terms.

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